2d Ising Model

Computational Physics Assignment

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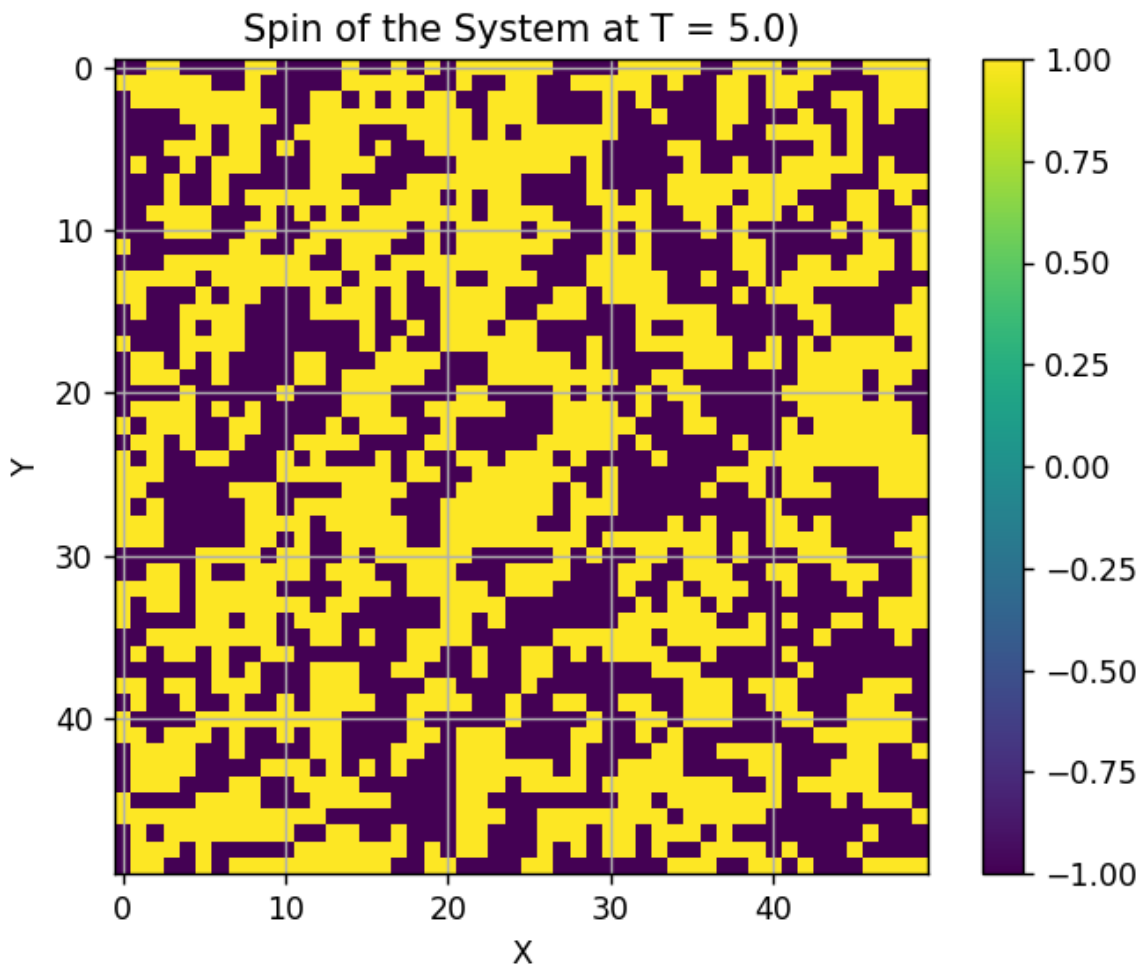
Code

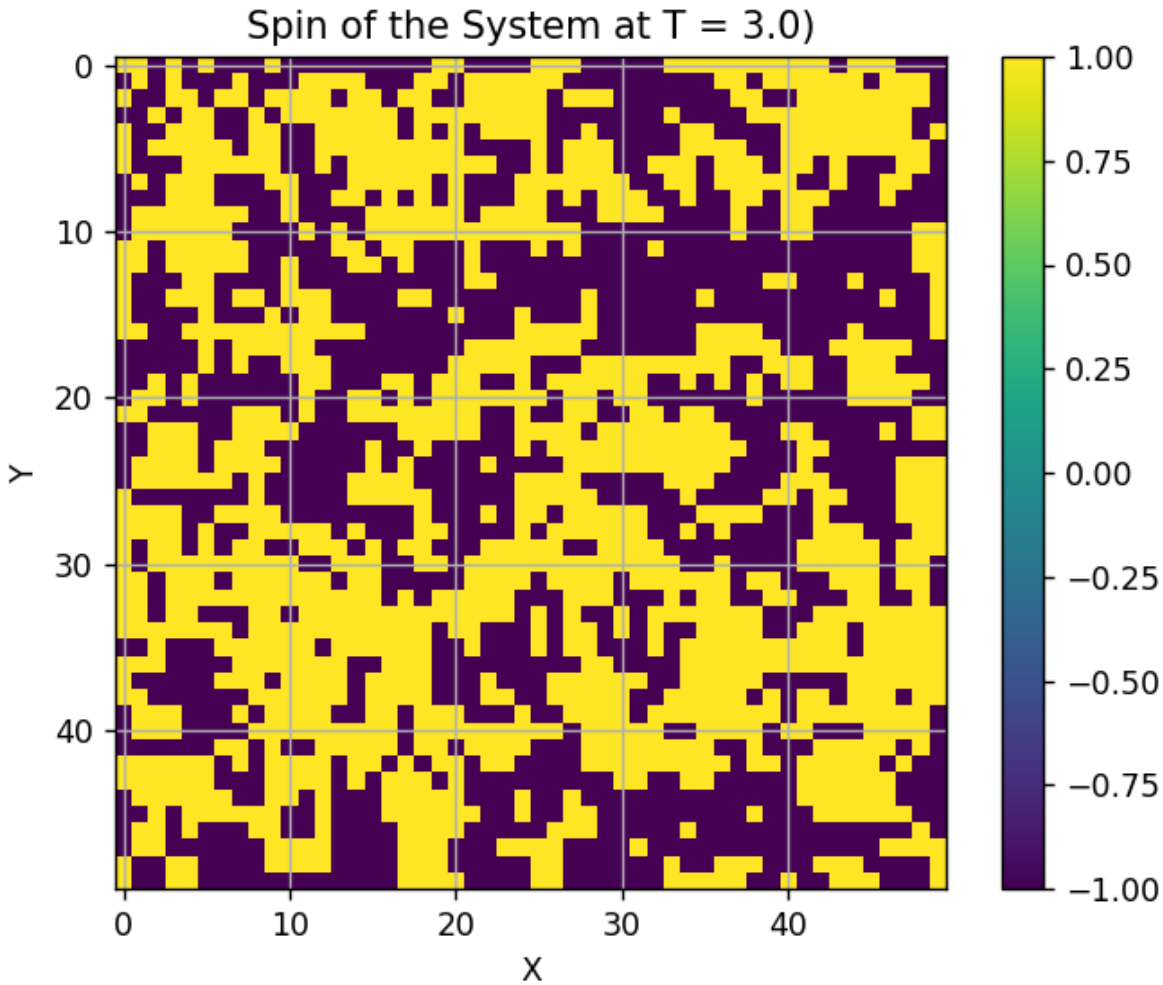
```
import numpy as np
import matplotlib.pyplot as plt
from scipy.signal import convolve2d

def initialize_spins(Lx, Ly):
    '''Create a 2d array of spin'''
    return np.random.choice([1,-1], size=(Lx,Ly))

def calculate_energy(spins, J):
    '''Calcuation of Energy of the 2d lattice using periodic
boundary condition'''
    #Multiplying the spins of nearest spin states
    kernel = np.array([
        [0, 1, 0],
        [1, 0, 1],
        [0, 1, 0]
    ])
    return -J * np.sum(spins*convolve2d(spins, kernel ,
mode="same", boundary="wrap")) / 2

def metropolis_step(spins, B, J):
    '''Perform single metropolis update on 2d lattice at random
location'''

    Lx, Ly = spins.shape

    for _ in range(Lx*Ly):
        i = np.random.randint(Lx)
        j = np.random.randint(Ly)
```

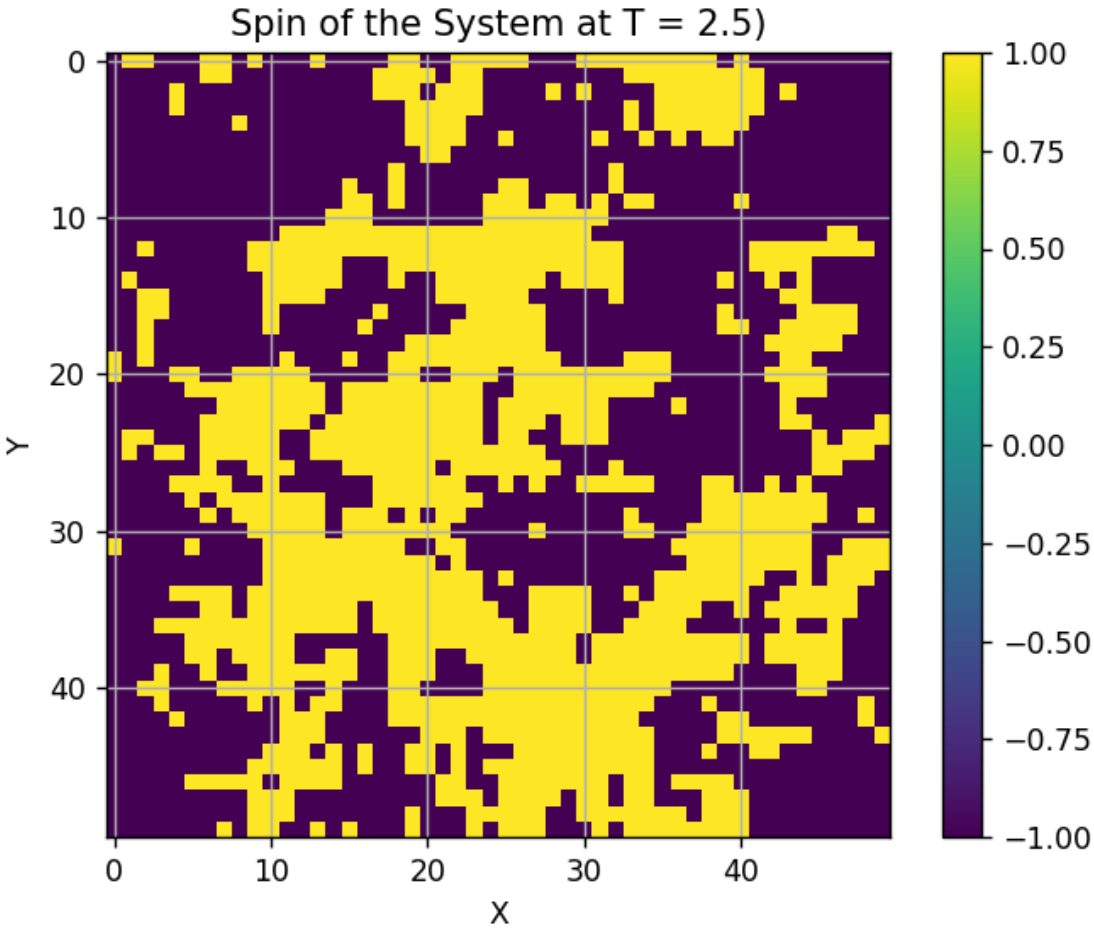
```
        delta_E = 2 * J * spins[i, j] * (spins[(i-1)%Lx, j] +
spins[(i+1)%Lx, j] + spins[i, (j+1)%Ly] + spins[i, (j-1)%Ly])
        # %Lx and %Ly is for periodic boundary condition

        if delta_E <= 0 or np.random.rand() < np.exp(- B *
delta_E):
            spins[i, j] *= -1
    return spins

def simulate_ising_2d(Lx, Ly, J, T, n_steps):
    """Run the 1D Ising model simulation."""
    B = 1.0 / T      # B --> beta
    spins = initialize_spins(Lx, Ly)
    magnetizations = []
    energies = []

    for step in range(n_steps):
        spins = metropolis_step(spins, B, J)
        magnetization = np.mean(spins)
        magnetizations.append(magnetization)
        energy = calculate_energy(spins, J)
        energies.append(energy)

    return spins, magnetizations, energies

def main():

    #Parameters
    L = 50            # Size of the the Square lattice
    T = 2.0           # Temperature in Kelvin
    J = 1.0           # Interaction Strength
    n_steps = 500    # Number of steps

    # Run simulation
    final_spins, mags, Es = simulate_ising_2d(L, L, J, T,
n_steps)
```

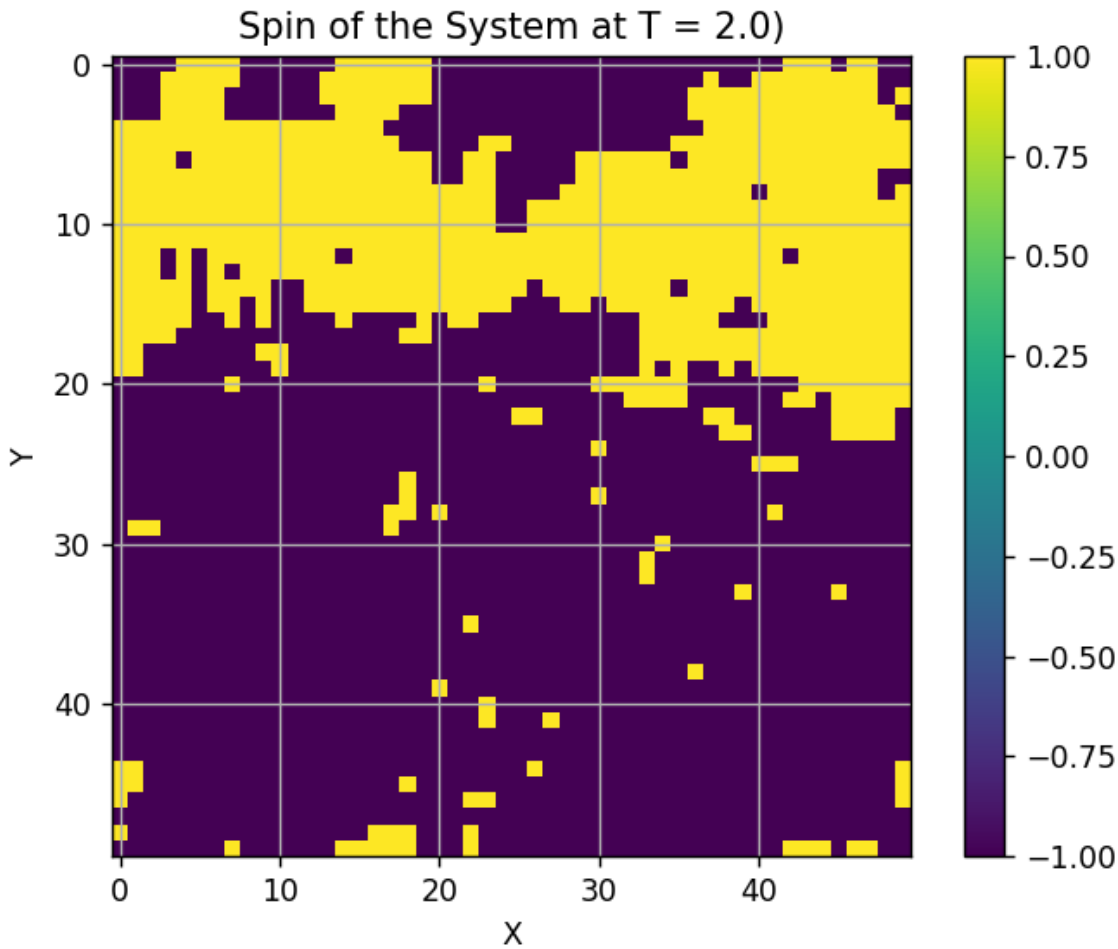
```python
    # Plot magnetization over time
    plt.figure(1)
    plt.plot(mags)
    plt.xlabel('Monte Carlo step')
    plt.ylabel('Magnetisation')
    plt.title(f'Magnetisation (Temperature = {T})')
    plt.grid(True)

    # Plot Energy over time
    plt.figure(2)
    plt.plot(Es)
    plt.xlabel('Monte Carlo step')
    plt.ylabel('E')
    plt.title(f'Energy Fluctuation (Temperature = {T})')
    plt.grid(True)

    # Plot Energy over time
    plt.figure(3)
    plt.imshow(final_spins)
    plt.xlabel('X')
    plt.ylabel('Y')
    plt.title(f'Spin of the System at T = {T})')
    plt.colorbar()
    plt.grid(True)
    plt.show()


if __name__ == "__main__":
    main()
```

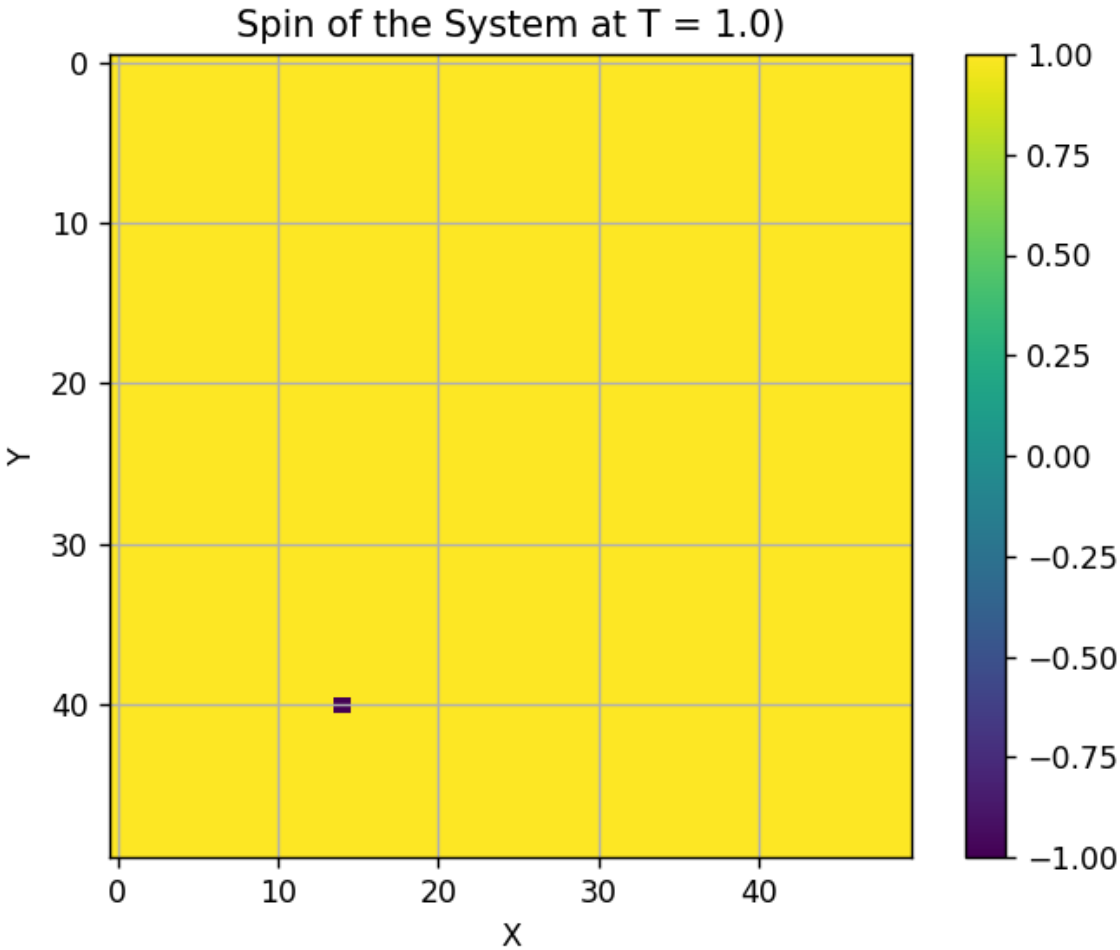
Question A


Spin of the System at T = 5.0)
0
10
20
30
40
Y
0
10
20
30
40
X
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00


Spin of the System at T = 3.0)
0
10
20
30
40
Y
0
10
20
30
40
X
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00


Spin of the System at T = 2.5)
0
10
20
30
40
Y
0
10
20
30
40
X
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00


Spin of the System at T = 2.0)
0
10
20
30
40
Y
0
10
20
30
40
X
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00

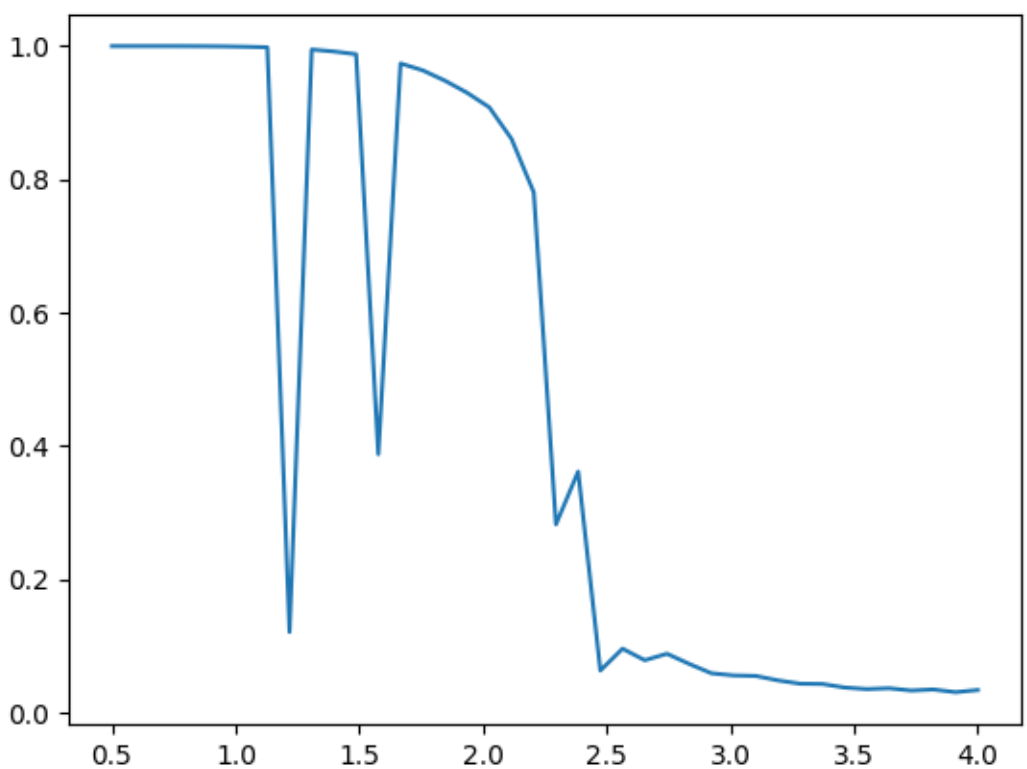


Question B

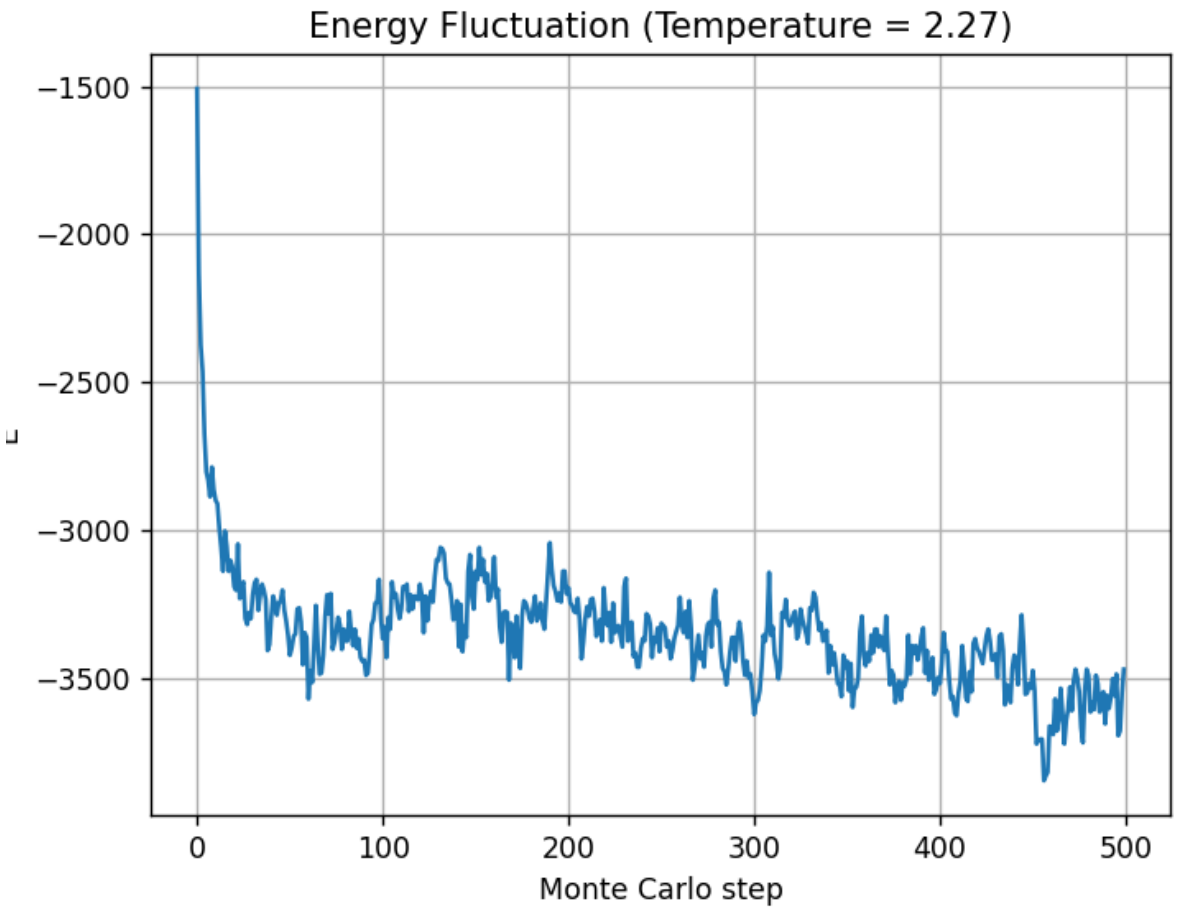
```
import numpy as np
import matplotlib.pyplot as plt
from Ising_Model_2d import simulate_ising_2d

def Mag_Vs_T():
    T = np.linspace(0.5,4, 20)
    M = np.zeros_like(T)
    J = 1.0
    N = 500
    Lx , Ly = (50,50)
```

```
    for i in range(len(T)):
        spins, mags, Es = simulate_ising_2d(Lx, Ly, J, T[i], N)

        M[i] = np.mean(np.abs(mags[0.5*len(mags):]))

    return T, M

T, M = Mag_Vs_T()

plt.plot(T, M)
plt.show()
```

Graph of Magnetization Vs Temperature is shown below:



Question C



The critical temperature (Tc)=2.3K in the 2D Ising model marks the transition between ordered and disordered phases. Below Tc, the system exhibits ferromagnetic behavior with aligned spins and spontaneous magnetization. As the temperature approaches Tcfrom below, the magnetization decreases sharply, vanishing entirely above Tc where the system becomes paramagnetic with randomly oriented spins. At Tc itself, the system displays scale-invariant fluctuations, meaning spin domains of all sizes appear, reflecting the divergence of the correlation length in the thermodynamic limit.

Energy fluctuations also vary significantly with temperature. In the ordered phase energy fluctuations are small because most spins are aligned, minimizing changes in the system's total energy. Near Tc, these fluctuations become large and pronounced due to the coexistence of ordered and disordered regions. Above Tc, energy fluctuations diminish again as the system settles into a fully disordered state with no preferred spin alignment. The behavior of these fluctuations near Tc is a hallmark of the phase transition, revealing the system's critical dynamics without requiring explicit calculation of specific heat.